

# NLP Lab: Autogen

Progress Update 05.07.24

# What was done

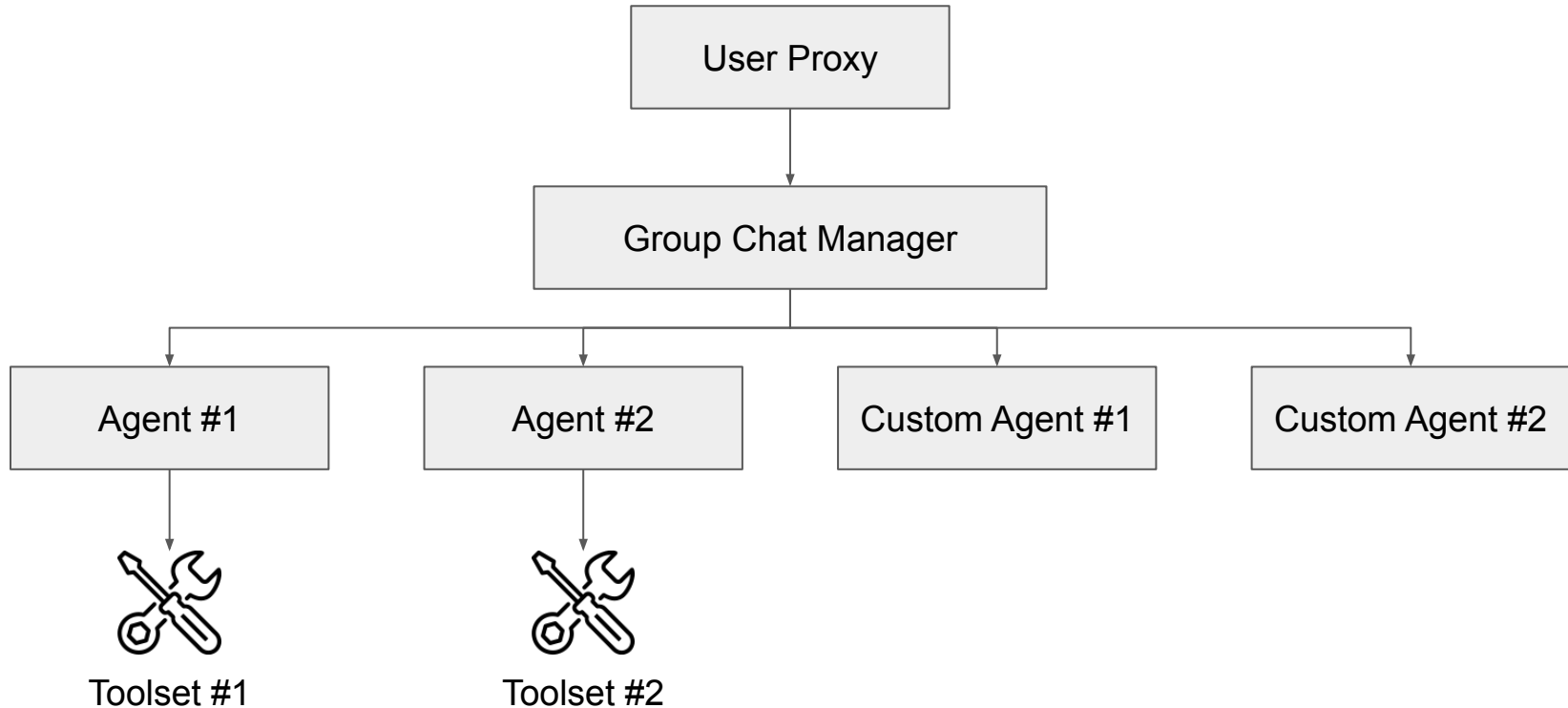
## Streamlit demos:

- Finished overarching agent demo for poster presentation
  - 6 agents with a delegated toolset for specific tasks
  - Option to add new agents with given name/prompt
  - Option to select participating agents
- Unified chat UI across almost all experiments

## Experiments:

- Created “SelectBenchmark” (120 tasks) to measure the accuracy of top agent
- Evaluation is planned until the final deadline

# Multiagent Architecture



# Demo time!

The demo is deployed at <http://130.162.48.238:8080/>

# SelectBenchmark

Contains 20 test cases each for calculator, coding, DALL-E, image\_retriever, pdf\_parser and the web\_retriever agent. Examples:

```
{
  "id": "TC005",
  "name": "Mixed Operations",
  "input": "What is (10 + 5) * 2 - 8?",
  "expected_agent": "calculator",
  "expected_result": 22,
  "success_criteria": [
    "Correct result (22) is returned"
  ]
},
{
  "id": "TC006",
  "name": "Decimal Addition",
  "input": "Add 3.14 and 2.86",
  "expected_agent": "calculator",
  "expected_result": 6,
  "success_criteria": [
    "Correct result (6) is returned"
  ]
},
{
  "id": "TC007",
  "name": "Negative Number Subtraction",
  "input": "What's the result of 5 minus -3?",
  "expected_agent": "calculator",
  "expected_result": 8,
  "success_criteria": [
    "Correct result (8) is returned"
  ]
},
}
```

Calculator

```
{
  "id": "TC001",
  "name": "Simple Object Generation",
  "input": "Generate a close-up shot of a red apple",
  "expected_agent": "dalle",
  "success_criteria": [
    "Image contains a clearly recognizable red apple",
    "Image is of high quality and resolution",
    "Image is a close-up shot"
  ]
},
{
  "id": "TC002",
  "name": "Complex Scene Generation",
  "input": "Create an image of a futuristic city skyline at night",
  "expected_agent": "dalle",
  "success_criteria": [
    "Image depicts a city skyline with futuristic elements",
    "Night-time atmosphere is conveyed",
    "Image shows detail and creativity in architectural designs"
  ]
},
{
  "id": "TC003",
  "name": "Abstract Concept Visualization",
  "input": "Visualize the concept of time as an abstract image",
  "expected_agent": "dalle",
  "success_criteria": [
    "Image presents an abstract representation related to time",
    "Creative interpretation of the concept is evident",
    "Image evokes a sense of time passing or movement"
  ]
},
}
```

DALL-E

```
{
  "id": "TC006",
  "name": "Art Style Analysis",
  "input": "What style of painting is this? Describe its key features.",
  "path_url": "https://images.unsplash.com/photo-1615184697985-c9bde1b07da7?q=80&w=1974&auto",
  "expected_agent": "image_retriever",
  "success_criteria": [
    "Identifies the style of painting (abstract art)",
    "Describes key features and techniques visible in the artwork"
  ]
},
{
  "id": "TC007",
  "name": "Vehicle Identification",
  "input": "What type of vehicle is this and what era is it from?",
  "path_url": "https://images.unsplash.com/photo-1635785657876-96567c4027f3?q=80&w=2070&auto",
  "expected_agent": "image_retriever",
  "success_criteria": [
    "Trabant produced in east germany",
    "Provides an estimate of the era or decade the vehicle is from (produced from 1957 to 1990)"
  ]
},
{
  "id": "TC008",
  "name": "Sports Scene Analysis",
  "input": "What sport is being played here and what's happening in the image?",
  "path_url": "https://images.unsplash.com/photo-1587329310686-91414b8e3cb7",
  "expected_agent": "image_retriever",
  "success_criteria": [
    "Identifies the sport being played (ball sports) and the specific sport (football)",
    "Describes the specific action or moment captured in the image (ball hitting the goal)"
  ]
},
}
```

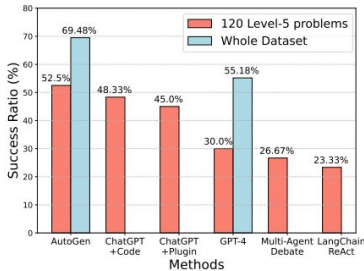
image\_retriever

# What is left to do

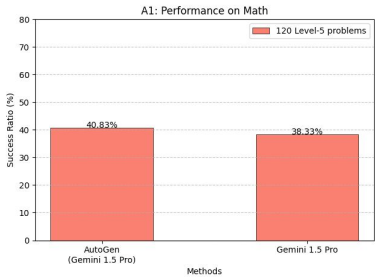
- Measure the accuracy of the parent agent (and other kinds of parents) on SelectBenchmark
- Documentation
- Poster by end of next week
  
- More possible experiments & benchmark
- Conduct potential experiments & do comparisons
- Extend the experiments type till the final ddl, if time allows (planning, reasoning, etc.)

# Math

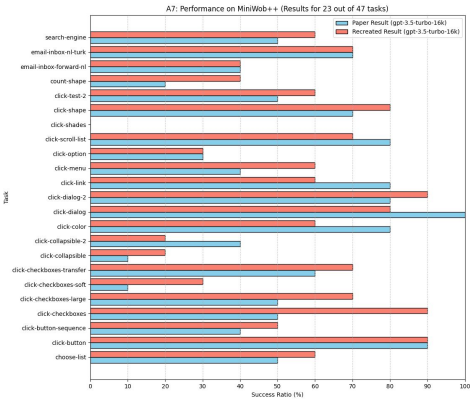
Original:



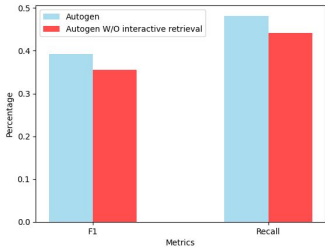
Ours:



# Online Decision Making

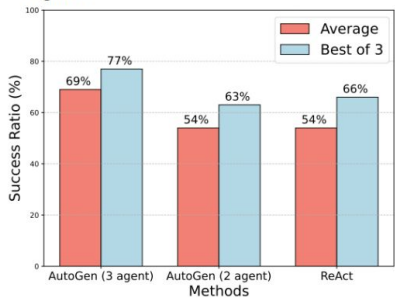


# RAG

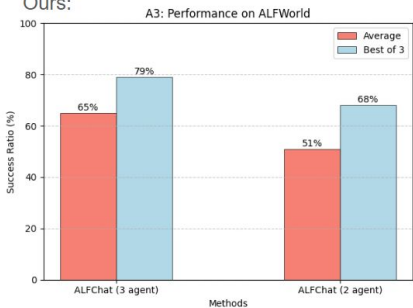


# ALFWorld

Original:



Ours:



# Dynamic Group Chat

| Model         | Two Agent | Group Chat |
|---------------|-----------|------------|
| GPT-3.5-Turbo | 6         | 8          |
| GPT-4         | 9         | 7          |

# Multi Agent Coding

| Dataset      | netflow      | facility     | tsp          | coffee       | diet         |
|--------------|--------------|--------------|--------------|--------------|--------------|
| Saving Ratio | 3.14x (0.65) | 3.14x (0.64) | 4.88x (1.71) | 3.38x (0.86) | 3.03x (0.31) |

|              |              |             |              |
|--------------|--------------|-------------|--------------|
| Dataset      | tsp          | coffee      | diet         |
| Saving Ratio | 4.65x (1.58) | 3.5x (0.90) | 3.02x (0.31) |